PG No. 15 D_{4h} $4/mmm$ [tetragonal] (internal axial quadrupole)

* Harmonics for rank 0

$\vec{\mathbb{G}}_0^{(2,2)}[g](A_{1u})$

** symmetry

1

** expression

$$-\frac{\sqrt{5}G_u\left(x^2+y^2-2z^2\right)}{10}+\frac{\sqrt{15}G_v\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{15}G_{xy}xy}{5}+\frac{\sqrt{15}G_{xz}xz}{5}+\frac{\sqrt{15}G_{yz}yz}{5}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(2,0)}[g](A_{2g})$

** symmetry

z

** expression

$$\frac{\sqrt{10}G_uz}{5}+\frac{\sqrt{30}G_{xz}x}{10}+\frac{\sqrt{30}G_{yz}y}{10}$$

$\vec{\mathbb{G}}_1^{(2,2)}[g](A_{2g})$

** symmetry

z

** expression

$$-\frac{3\sqrt{35}G_uz\left(3x^2+3y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_vz\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}G_{xy}xyz}{7}-\frac{\sqrt{105}G_{xz}x\left(x^2+y^2-4z^2\right)}{35}-\frac{\sqrt{105}G_{yz}y\left(x^2+y^2-4z^2\right)}{35}$$

$\vec{\mathbb{G}}_{1,1}^{(2,0)}[g](E_g)$, $\vec{\mathbb{G}}_{1,2}^{(2,0)}[g](E_g)$

** symmetry

x

$-y$

** expression

$$-\frac{\sqrt{10}G_ux}{10}+\frac{\sqrt{30}G_vx}{10}+\frac{\sqrt{30}G_{xy}y}{10}+\frac{\sqrt{30}G_{xz}z}{10}$$

$$\frac{\sqrt{10}G_uy}{10}+\frac{\sqrt{30}G_vy}{10}-\frac{\sqrt{30}G_{xy}x}{10}-\frac{\sqrt{30}G_{yz}z}{10}$$

$\vec{\mathbb{G}}_{1,1}^{(2,2)}[g](E_g)$, $\vec{\mathbb{G}}_{1,2}^{(2,2)}[g](E_g)$

** symmetry

x

$-y$

** expression

$$-\frac{3\sqrt{35}G_ux\left(x^2+y^2-4z^2\right)}{70}+\frac{\sqrt{105}G_vx\left(3x^2-7y^2-2z^2\right)}{70}+\frac{\sqrt{105}G_{xy}y\left(4x^2-y^2-z^2\right)}{35}+\frac{\sqrt{105}G_{xz}z\left(4x^2-y^2-z^2\right)}{35}+\frac{\sqrt{105}G_{yz}xyz}{7}$$

$$\frac{3\sqrt{35}G_uy\left(x^2+y^2-4z^2\right)}{70}-\frac{\sqrt{105}G_vy\left(7x^2-3y^2+2z^2\right)}{70}+\frac{\sqrt{105}G_{xy}x\left(x^2-4y^2+z^2\right)}{35}-\frac{\sqrt{105}G_{xz}xyz}{7}+\frac{\sqrt{105}G_{yz}z\left(x^2-4y^2+z^2\right)}{35}$$

$\vec{\mathbb{Q}}_1^{(2,1)}[g](A_{2u})$

** symmetry

z

** expression

$$\frac{\sqrt{30}G_vxy}{5}-\frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{30}G_{xz}yz}{10}-\frac{\sqrt{30}G_{yz}xz}{10}$$

$\vec{\mathbb{Q}}_{1,1}^{(2,1)}[g](E_u)$, $\vec{\mathbb{Q}}_{1,2}^{(2,1)}[g](E_u)$

** symmetry

x

y

** expression

$$-\frac{3\sqrt{10}G_u yz}{10}-\frac{\sqrt{30}G_v yz}{10}+\frac{\sqrt{30}G_{xy}xz}{10}-\frac{\sqrt{30}G_{xz}xy}{10}-\frac{\sqrt{30}G_{yz}\left(y-z\right)\left(y+z\right)}{10}$$

$$\frac{3\sqrt{10}G_u xz}{10}-\frac{\sqrt{30}G_v xz}{10}-\frac{\sqrt{30}G_{xy}yz}{10}+\frac{\sqrt{30}G_{xz}\left(x-z\right)\left(x+z\right)}{10}+\frac{\sqrt{30}G_{yz}xy}{10}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](A_{1g})$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{2}G_{xz}y}{2}-\frac{\sqrt{2}G_{yz}x}{2}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](A_{1g})$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{5\sqrt{42}G_v xyz}{14}-\frac{5\sqrt{42}G_{xy}z\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{42}G_{xz}y\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{42}G_{yz}x\left(x^2+y^2-4z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](B_{1g})$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{6}G_{xy}z}{3}-\frac{\sqrt{6}G_{xz}y}{6}-\frac{\sqrt{6}G_{yz}x}{6}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](B_{1g})$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{5\sqrt{42}G_u xyz}{14}+\frac{\sqrt{14}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{28}-\frac{\sqrt{14}G_{xz}y\left(9x^2-y^2-6z^2\right)}{28}+\frac{\sqrt{14}G_{yz}x\left(x^2-9y^2+6z^2\right)}{28}$$

$\vec{\mathbb{Q}}_2^{(2,-1)}[g](B_{2g})$

** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{6}G_v z}{3}+\frac{\sqrt{6}G_{xz}x}{6}-\frac{\sqrt{6}G_{yz}y}{6}$$

$\vec{\mathbb{Q}}_2^{(2,1)}[g](B_{2g})$

** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{5\sqrt{42}G_u z\left(x-y\right)\left(x+y\right)}{28}-\frac{\sqrt{14}G_v z\left(3x^2+3y^2-2z^2\right)}{28}+\frac{\sqrt{14}G_{xz}x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{14}G_{yz}y\left(3x^2-2y^2+3z^2\right)}{14}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,-1)}[g](E_g)$, $\vec{\mathbb{Q}}_{2,2}^{(2,-1)}[g](E_g)$

** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{2}G_u x}{2} + \frac{\sqrt{6}G_v x}{6} + \frac{\sqrt{6}G_{xy} y}{6} - \frac{\sqrt{6}G_{xz} z}{6}$$

$$-\frac{\sqrt{2}G_u y}{2} + \frac{\sqrt{6}G_v y}{6} - \frac{\sqrt{6}G_{xy} x}{6} + \frac{\sqrt{6}G_{yz} z}{6}$$

$\vec{\mathbb{Q}}_{2,1}^{(2,1)}[g](E_g)$, $\vec{\mathbb{Q}}_{2,2}^{(2,1)}[g](E_g)$
** symmetry

$$\sqrt{3}yz$$

$$\sqrt{3}xz$$

** expression

$$-\frac{\sqrt{42}G_u x\left(x^2+y^2-4z^2\right)}{28} - \frac{\sqrt{14}G_v x\left(x^2-9y^2+6z^2\right)}{28} - \frac{\sqrt{14}G_{xy} y\left(3x^2-2y^2+3z^2\right)}{14} + \frac{\sqrt{14}G_{xz} z\left(3x^2+3y^2-2z^2\right)}{14}$$

$$\frac{\sqrt{42}G_u y\left(x^2+y^2-4z^2\right)}{28} + \frac{\sqrt{14}G_v y\left(9x^2-y^2-6z^2\right)}{28} - \frac{\sqrt{14}G_{xy} x\left(2x^2-3y^2-3z^2\right)}{14} - \frac{\sqrt{14}G_{yz} z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$G_u$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$-\frac{\sqrt{14}G_u\left(x^2+y^2-2z^2\right)}{14} - \frac{\sqrt{42}G_v(x-y)(x+y)}{14} - \frac{\sqrt{42}G_{xy}xy}{7} + \frac{\sqrt{42}G_{xz}xz}{14} + \frac{\sqrt{42}G_{yz}yz}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](A_{1u})$
** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{14}G_u\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{56} - \frac{5\sqrt{42}G_v(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{168}$$
$$- \frac{5\sqrt{42}G_{xy}xy\left(x^2+y^2-6z^2\right)}{84} - \frac{5\sqrt{42}G_{xz}xz\left(3x^2+3y^2-4z^2\right)}{84} - \frac{5\sqrt{42}G_{yz}yz\left(3x^2+3y^2-4z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](B_{1u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$G_v$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](B_{1u})$
** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{42}G_u(x-y)(x+y)}{14} + \frac{\sqrt{14}G_v\left(x^2+y^2-2z^2\right)}{14} + \frac{3\sqrt{14}G_{xz}xz}{14} - \frac{3\sqrt{14}G_{yz}yz}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](B_{1u})$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{5\sqrt{42}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168} + \frac{\sqrt{14}G_v\left(19x^4-102x^2y^2-12x^2z^2+19y^4-12y^2z^2+4z^4\right)}{168}$$
$$+\frac{5\sqrt{14}G_{xy}xy\left(x-y\right)\left(x+y\right)}{12} + \frac{5\sqrt{14}G_{xz}xz\left(5x^2-9y^2-2z^2\right)}{84} + \frac{5\sqrt{14}G_{yz}yz\left(9x^2-5y^2+2z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(2,-2)}[g](B_{2u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$G_{xy}$$

$\vec{\mathbb{G}}_2^{(2,0)}[g](B_{2u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}G_uxy}{7} + \frac{\sqrt{14}G_{xy}\left(x^2+y^2-2z^2\right)}{14} + \frac{3\sqrt{14}G_{xz}yz}{14} + \frac{3\sqrt{14}G_{yz}xz}{14}$$

$\vec{\mathbb{G}}_2^{(2,2)}[g](B_{2u})$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{5\sqrt{42}G_uxy\left(x^2+y^2-6z^2\right)}{84} + \frac{5\sqrt{14}G_vxy\left(x-y\right)\left(x+y\right)}{12} - \frac{\sqrt{14}G_{xy}\left(4x^4-27x^2y^2+3x^2z^2+4y^4+3y^2z^2-z^4\right)}{42}$$
$$+\frac{5\sqrt{14}G_{xz}yz\left(6x^2-y^2-z^2\right)}{42} - \frac{5\sqrt{14}G_{yz}xz\left(x^2-6y^2+z^2\right)}{42}$$

$\vec{\mathbb{G}}_{2,1}^{(2,-2)}[g](E_u)$, $\vec{\mathbb{G}}_{2,2}^{(2,-2)}[g](E_u)$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$G_{yz}$$

$$-G_{xz}$$

$\vec{\mathbb{G}}_{2,1}^{(2,0)}[g](E_u)$, $\vec{\mathbb{G}}_{2,2}^{(2,0)}[g](E_u)$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$\frac{\sqrt{42}G_uyz}{14} - \frac{3\sqrt{14}G_vyz}{14} + \frac{3\sqrt{14}G_{xy}xz}{14} + \frac{3\sqrt{14}G_{xz}xy}{14} - \frac{\sqrt{14}G_{yz}\left(2x^2-y^2-z^2\right)}{14}$$

$$-\frac{\sqrt{42}G_uxz}{14} - \frac{3\sqrt{14}G_vxz}{14} - \frac{3\sqrt{14}G_{xy}yz}{14} - \frac{\sqrt{14}G_{xz}\left(x^2-2y^2+z^2\right)}{14} - \frac{3\sqrt{14}G_{yz}xy}{14}$$

$\vec{\mathbb{G}}_{2,1}^{(2,2)}[g](E_u)$, $\vec{\mathbb{G}}_{2,2}^{(2,2)}[g](E_u)$
** symmetry

$$\sqrt{3}yz$$

$$-\sqrt{3}xz$$

** expression

$$-\frac{5\sqrt{42}G_u yz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{14}G_v yz\left(9x^2-5y^2+2z^2\right)}{84}-\frac{5\sqrt{14}G_{xy}xz\left(x^2-6y^2+z^2\right)}{42}$$
$$-\frac{5\sqrt{14}G_{xz}xy\left(x^2+y^2-6z^2\right)}{42}+\frac{\sqrt{14}G_{yz}\left(x^4-3x^2y^2-3x^2z^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

$$\frac{5\sqrt{42}G_u xz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{5\sqrt{14}G_v xz\left(5x^2-9y^2-2z^2\right)}{84}-\frac{5\sqrt{14}G_{xy}yz\left(6x^2-y^2-z^2\right)}{42}$$
$$+\frac{\sqrt{14}G_{xz}\left(4x^4+3x^2y^2-27x^2z^2-y^4+3y^2z^2+4z^4\right)}{42}+\frac{5\sqrt{14}G_{yz}xy\left(x^2+y^2-6z^2\right)}{42}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_3^{(2,-2)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{15}G_u z}{5}-\frac{\sqrt{5}G_{xz}x}{5}-\frac{\sqrt{5}G_{yz}y}{5}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_u z\left(3x^2+3y^2-2z^2\right)}{15}-\frac{\sqrt{5}G_v z\left(x-y\right)\left(x+y\right)}{2}-\sqrt{5}G_{xy}xyz-\frac{\sqrt{5}G_{xz}x\left(x^2+y^2-4z^2\right)}{20}-\frac{\sqrt{5}G_{yz}y\left(x^2+y^2-4z^2\right)}{20}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](A_{2g})$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{330}G_u z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}-\frac{7\sqrt{110}G_v z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}-\frac{7\sqrt{110}G_{xy}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{\sqrt{110}G_{xz}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}+\frac{\sqrt{110}G_{yz}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](B_{1g})$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{3}G_{xy}z}{3}+\frac{\sqrt{3}G_{xz}y}{3}+\frac{\sqrt{3}G_{yz}x}{3}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](B_{1g})$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{3}G_{xy}z\left(3x^2+3y^2-2z^2\right)}{6}+\frac{\sqrt{3}G_{xz}y\left(3x^2-2y^2+3z^2\right)}{6}-\frac{\sqrt{3}G_{yz}x\left(2x^2-3y^2-3z^2\right)}{6}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](B_{1g})$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{21\sqrt{22}G_u xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{21\sqrt{66}G_v xyz\left(x-y\right)\left(x+y\right)}{44}-\frac{\sqrt{66}G_{xy}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{66}$$
$$-\frac{\sqrt{66}G_{xz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{66}+\frac{\sqrt{66}G_{yz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{66}$$

$\vec{\mathbb{G}}_3^{(2,-2)}[g](B_{2g})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_vz}{3} + \frac{\sqrt{3}G_{xz}x}{3} - \frac{\sqrt{3}G_{yz}y}{3}$$

$\vec{\mathbb{G}}_3^{(2,0)}[g](B_{2g})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_vz\left(3x^2+3y^2-2z^2\right)}{6} + \frac{\sqrt{3}G_{xz}x\left(x^2-9y^2+6z^2\right)}{12} + \frac{\sqrt{3}G_{yz}y\left(9x^2-y^2-6z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(2,2)}[g](B_{2g})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{21\sqrt{22}G_uz\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88} + \frac{\sqrt{66}G_vz\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{264} + \frac{21\sqrt{66}G_{xy}xyz\left(x-y\right)\left(x+y\right)}{44}$$
$$-\frac{\sqrt{66}G_{xz}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{132} - \frac{\sqrt{66}G_{yz}y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{132}$$

$\vec{\mathbb{G}}_{3,1}^{(2,-2)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(2,-2)}[g](E_g,1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{15}G_ux}{10} + \frac{3\sqrt{5}G_vx}{10} - \frac{\sqrt{5}G_{xy}y}{5} - \frac{\sqrt{5}G_{xz}z}{5}$$

$$\frac{\sqrt{15}G_uy}{10} + \frac{3\sqrt{5}G_vy}{10} + \frac{\sqrt{5}G_{xy}x}{5} + \frac{\sqrt{5}G_{yz}z}{5}$$

$\vec{\mathbb{G}}_{3,1}^{(2,-2)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(2,-2)}[g](E_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{G_ux}{2} - \frac{\sqrt{3}G_vx}{6} + \frac{\sqrt{3}G_{xy}y}{3} - \frac{\sqrt{3}G_{xz}z}{3}$$

$$\frac{G_uy}{2} - \frac{\sqrt{3}G_vy}{6} - \frac{\sqrt{3}G_{xy}x}{3} + \frac{\sqrt{3}G_{yz}z}{3}$$

$\vec{\mathbb{G}}_{3,1}^{(2,0)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(2,0)}[g](E_g,1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$\frac{3x^2y}{2} - y^3 + \frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{15}G_u x\left(4x^2-21y^2+9z^2\right)}{60}+\frac{\sqrt{5}G_v x\left(4x^2-y^2-11z^2\right)}{20}+\frac{\sqrt{5}G_{xy} y\left(4x^2-y^2-z^2\right)}{20}+\frac{\sqrt{5}G_{xz} z\left(4x^2-y^2-z^2\right)}{20}-\sqrt{5}G_{yz}xyz$$

$$-\frac{\sqrt{15}G_u y\left(21x^2-4y^2-9z^2\right)}{60}-\frac{\sqrt{5}G_v y\left(x^2-4y^2+11z^2\right)}{20}+\frac{\sqrt{5}G_{xy} x\left(x^2-4y^2+z^2\right)}{20}+\sqrt{5}G_{xz}xyz+\frac{\sqrt{5}G_{yz} z\left(x^2-4y^2+z^2\right)}{20}$$

$\vec{\mathbb{G}}_{3,1}^{(2,0)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(2,0)}[g](E_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{G_u x\left(2x^2-3y^2-3z^2\right)}{4}+\frac{\sqrt{3}G_v x\left(2x^2-3y^2-3z^2\right)}{12}+\frac{\sqrt{3}G_{xy} y\left(6x^2+y^2-9z^2\right)}{12}-\frac{\sqrt{3}G_{xz} z\left(6x^2-9y^2+z^2\right)}{12}$$

$$\frac{G_u y\left(3x^2-2y^2+3z^2\right)}{4}-\frac{\sqrt{3}G_v y\left(3x^2-2y^2+3z^2\right)}{12}-\frac{\sqrt{3}G_{xy} x\left(x^2+6y^2-9z^2\right)}{12}-\frac{\sqrt{3}G_{yz} z\left(9x^2-6y^2-z^2\right)}{12}$$

$\vec{\mathbb{G}}_{3,1}^{(2,2)}[g](E_g,1)$, $\vec{\mathbb{G}}_{3,2}^{(2,2)}[g](E_g,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$\frac{3x^2y}{2}-y^3+\frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{330}G_u x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{264}+\frac{\sqrt{110}G_v x\left(4x^4-27x^2y^2-13x^2z^2+11y^4+15y^2z^2+4z^4\right)}{88}$$
$$+\frac{\sqrt{110}G_{xy} y\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{44}+\frac{\sqrt{110}G_{xz} z\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{44}+\frac{7\sqrt{110}G_{yz}xyz\left(2x^2-y^2-z^2\right)}{44}$$

$$-\frac{\sqrt{330}G_u y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{264}+\frac{\sqrt{110}G_v y\left(11x^4-27x^2y^2+15x^2z^2+4y^4-13y^2z^2+4z^4\right)}{88}$$
$$-\frac{\sqrt{110}G_{xy} x\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{44}+\frac{7\sqrt{110}G_{xz}xyz\left(x^2-2y^2+z^2\right)}{44}-\frac{\sqrt{110}G_{yz} z\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{44}$$

$\vec{\mathbb{G}}_{3,1}^{(2,2)}[g](E_g,2)$, $\vec{\mathbb{G}}_{3,2}^{(2,2)}[g](E_g,2)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{\sqrt{22}G_u x\left(2x^4+11x^2y^2-31x^2z^2+9y^4-87y^2z^2+30z^4\right)}{88}-\frac{\sqrt{66}G_v x\left(2x^4-73x^2y^2+53x^2z^2+51y^4-87y^2z^2-12z^4\right)}{264}$$
$$-\frac{\sqrt{66}G_{xy} y\left(12x^4-46x^2y^2+66x^2z^2+5y^4-4y^2z^2-9z^4\right)}{132}$$
$$+\frac{\sqrt{66}G_{xz} z\left(12x^4+66x^2y^2-46x^2z^2-9y^4-4y^2z^2+5z^4\right)}{132}+\frac{21\sqrt{66}G_{yz}xyz\left(y-z\right)\left(y+z\right)}{44}$$

$$\frac{\sqrt{22}G_u y\left(9x^4+11x^2y^2-87x^2z^2+2y^4-31y^2z^2+30z^4\right)}{88}-\frac{\sqrt{66}G_v y\left(51x^4-73x^2y^2-87x^2z^2+2y^4+53y^2z^2-12z^4\right)}{264}$$
$$+\frac{\sqrt{66}G_{xy} x\left(5x^4-46x^2y^2-4x^2z^2+12y^4+66y^2z^2-9z^4\right)}{132}-\frac{21\sqrt{66}G_{xz}xyz\left(x-z\right)\left(x+z\right)}{44}$$
$$+\frac{\sqrt{66}G_{yz} z\left(9x^4-66x^2y^2+4x^2z^2-12y^4+46y^2z^2-5z^4\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{30}G_v xy}{10}+\frac{\sqrt{30}G_{xy}\left(x-y\right)\left(x+y\right)}{20}+\frac{\sqrt{30}G_{xz}yz}{5}-\frac{\sqrt{30}G_{yz}xz}{5}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](A_{2u})$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}G_v xy\left(x^2+y^2-6z^2\right)}{6}+\frac{\sqrt{15}G_{xy}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}-\frac{\sqrt{15}G_{xz}yz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{\sqrt{15}G_{yz}xz\left(3x^2+3y^2-4z^2\right)}{12}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](B_{1u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{6}G_u\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{2}G_v\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](B_{1u})$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}G_u\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{6}-\frac{G_v\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{6}$$
$$-\frac{7G_{xy}xy\left(x-y\right)\left(x+y\right)}{6}+\frac{7G_{xz}xz\left(x-z\right)\left(x+z\right)}{6}-\frac{7G_{yz}yz\left(y-z\right)\left(y+z\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(2,-1)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{6}G_u xy}{2}-\frac{\sqrt{2}G_{xy}\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{Q}}_3^{(2,1)}[g](B_{2u})$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}G_u xy\left(x^2+y^2-6z^2\right)}{3}+\frac{7G_v xy\left(x-y\right)\left(x+y\right)}{6}-\frac{G_{xy}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{12}$$
$$-\frac{7G_{xz}yz\left(3x^2+y^2-2z^2\right)}{12}-\frac{7G_{yz}xz\left(x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,-1)}[g](E_u,1)$, $\vec{\mathbb{Q}}_{3,2}^{(2,-1)}[g](E_u,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$\frac{3\sqrt{10}G_u yz}{20}+\frac{\sqrt{30}G_v yz}{20}+\frac{\sqrt{30}G_{xy}xz}{5}-\frac{\sqrt{30}G_{xz}xy}{5}+\frac{\sqrt{30}G_{yz}\left(y-z\right)\left(y+z\right)}{20}$$

$$-\frac{3\sqrt{10}G_u xz}{20}+\frac{\sqrt{30}G_v xz}{20}-\frac{\sqrt{30}G_{xy}yz}{5}-\frac{\sqrt{30}G_{xz}\left(x-z\right)\left(x+z\right)}{20}+\frac{\sqrt{30}G_{yz}xy}{5}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,-1)}[g](E_u,2)$, $\vec{\mathbb{Q}}_{3,2}^{(2,-1)}[g](E_u,2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{6}G_u yz}{4} - \frac{3\sqrt{2}G_v yz}{4} + \frac{\sqrt{2}G_{yz}\left(2x^2 - y^2 - z^2\right)}{4}$$

$$-\frac{\sqrt{6}G_u xz}{4} - \frac{3\sqrt{2}G_v xz}{4} + \frac{\sqrt{2}G_{xz}\left(x^2 - 2y^2 + z^2\right)}{4}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,1)}[g](E_u, 1)$, $\vec{\mathbb{Q}}_{3,2}^{(2,1)}[g](E_u, 1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$-\frac{\sqrt{5}G_u yz\left(6x^2 - y^2 - z^2\right)}{4} - \frac{\sqrt{15}G_v yz\left(6x^2 - y^2 - z^2\right)}{12} + \frac{\sqrt{15}G_{xy}xz\left(4x^2 - 3y^2 - 3z^2\right)}{12}$$
$$- \frac{\sqrt{15}G_{xz}xy\left(4x^2 - 3y^2 - 3z^2\right)}{12} - \frac{\sqrt{15}G_{yz}\left(y - z\right)\left(y + z\right)\left(6x^2 - y^2 - z^2\right)}{12}$$

$$-\frac{\sqrt{5}G_u xz\left(x^2 - 6y^2 + z^2\right)}{4} + \frac{\sqrt{15}G_v xz\left(x^2 - 6y^2 + z^2\right)}{12} + \frac{\sqrt{15}G_{xy}yz\left(3x^2 - 4y^2 + 3z^2\right)}{12}$$
$$- \frac{\sqrt{15}G_{xz}\left(x - z\right)\left(x + z\right)\left(x^2 - 6y^2 + z^2\right)}{12} - \frac{\sqrt{15}G_{yz}xy\left(3x^2 - 4y^2 + 3z^2\right)}{12}$$

$\vec{\mathbb{Q}}_{3,1}^{(2,1)}[g](E_u, 2)$, $\vec{\mathbb{Q}}_{3,2}^{(2,1)}[g](E_u, 2)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

$$\frac{\sqrt{15}x^2y}{2} - \frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{3}G_u yz\left(12x^2 - 9y^2 + 5z^2\right)}{12} - \frac{G_v yz\left(36x^2 + y^2 - 13z^2\right)}{12} + \frac{7G_{xy}xz\left(2x^2 - 3y^2 - z^2\right)}{12}$$
$$+ \frac{7G_{xz}xy\left(2x^2 - y^2 - 3z^2\right)}{12} - \frac{G_{yz}\left(4x^4 - 12x^2y^2 - 12x^2z^2 + 5y^4 - 18y^2z^2 + 5z^4\right)}{12}$$

$$\frac{\sqrt{3}G_u xz\left(9x^2 - 12y^2 - 5z^2\right)}{12} - \frac{G_v xz\left(x^2 + 36y^2 - 13z^2\right)}{12} + \frac{7G_{xy}yz\left(3x^2 - 2y^2 + z^2\right)}{12}$$
$$+ \frac{G_{xz}\left(5x^4 - 12x^2y^2 - 18x^2z^2 + 4y^4 - 12y^2z^2 + 5z^4\right)}{12} + \frac{7G_{yz}xy\left(x^2 - 2y^2 + 3z^2\right)}{12}$$